Lab Report 5: TurtleBot3 Simulation, SLAM, and Custom ROS 2 Nodes

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Course: Mobile Robotics (MCT-454L)

Workspace: ~/ros2_ws_imaan

Platform: Ubuntu Linux, ROS 2 Humble, Gazebo, RViz

1. Objective

The purpose of this lab was to transition from 2D kinematics (turtlesim) to 3D physics-based simulation. Key goals included:

- Simulating the **TurtleBot3 Burger** in a Gazebo environment.
- Implementing **SLAM (Simultaneous Localization and Mapping)** using the Cartographer algorithm.
- Visualizing sensor data (LiDAR, Odometry, TF) in **RViz**.
- Developing custom nodes to automate motion and parse complex odometry data.

2. Methodology & Implementation

Phase I: Environment & Simulation Setup

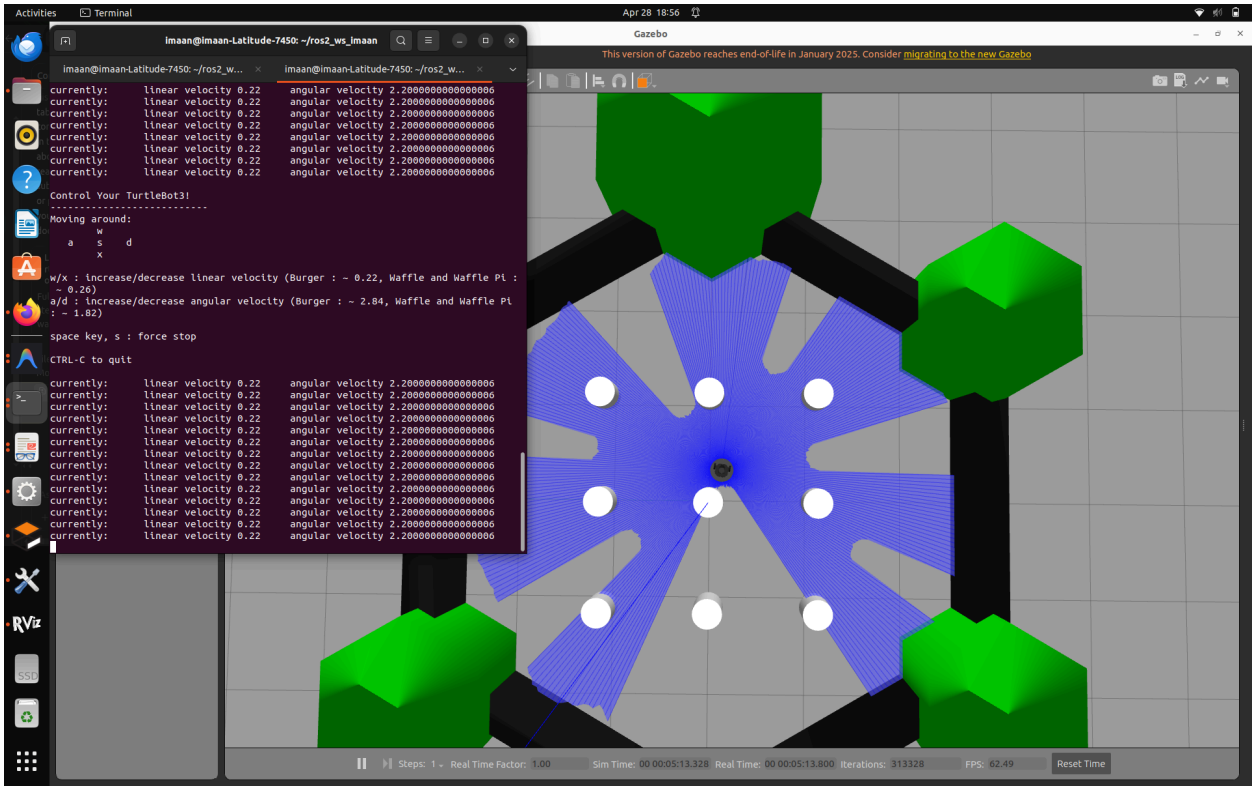
Before launching the simulator, the environment was configured for the TurtleBot3 Burger model.

- **Environment Configuration:**

Bash

```
source /opt/ros/humble/setup.bash
export TURTLEBOT3_MODEL=burger
```

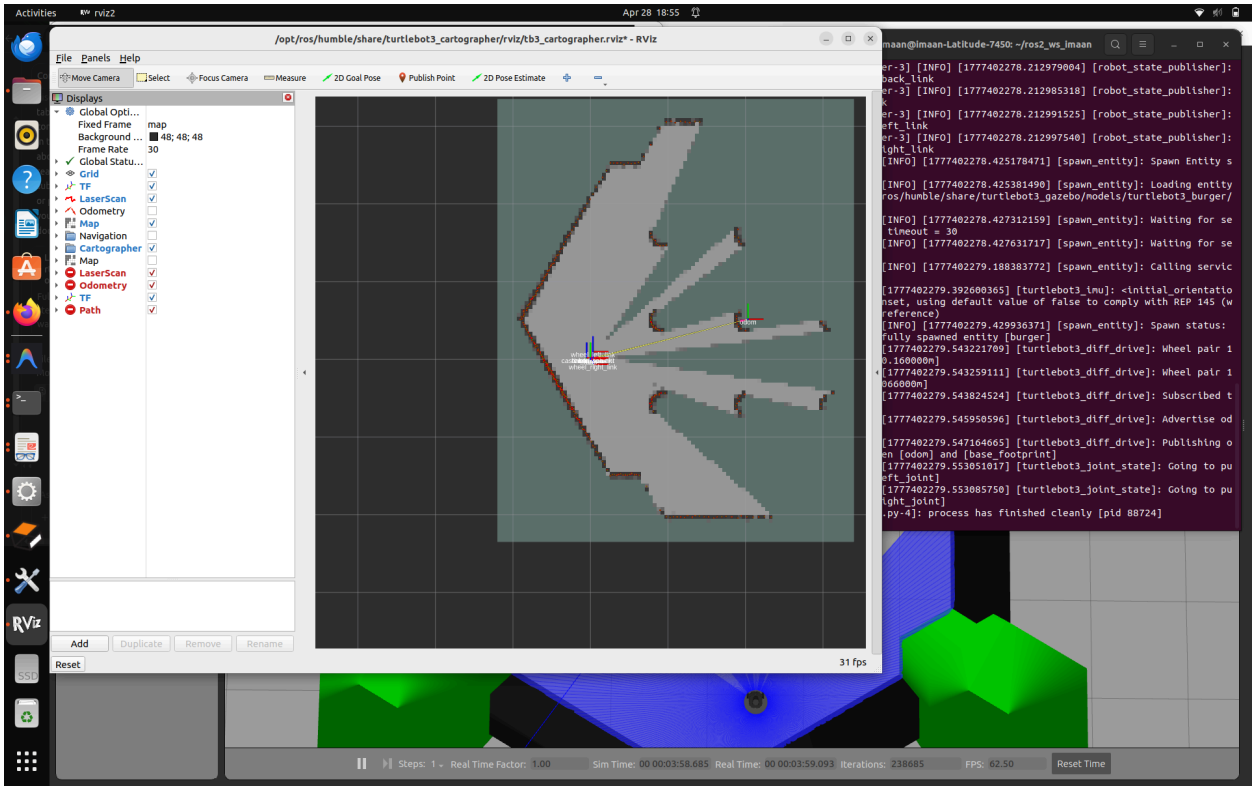
- **Gazebo Launch:** The world was initialized using: ros2 launch turtlebot3_gazebo turtlebot3_world.launch.py. This provided the physical simulation where the robot interacts with obstacles.



Phase II: SLAM and Visualization

Using **Cartographer**, I generated a real-time occupancy grid map.

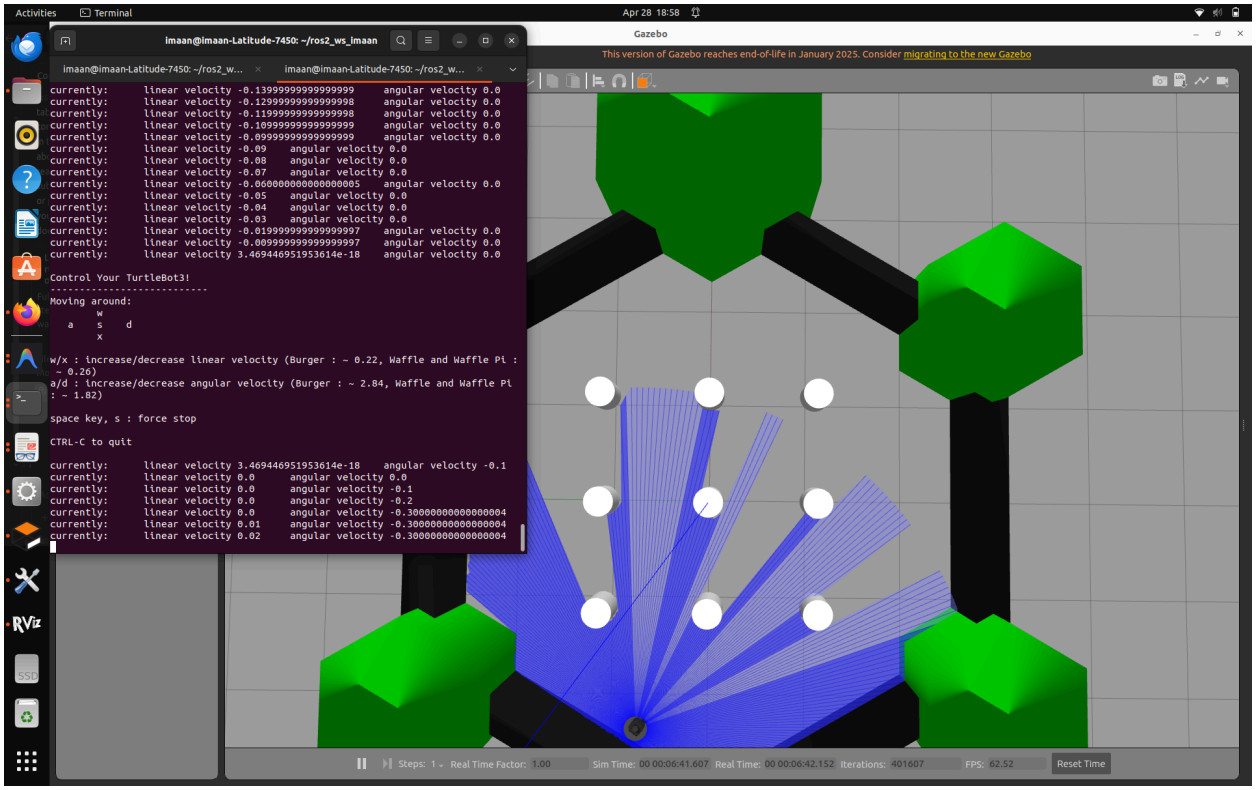
- **RViz Configuration:** I added plugins for **LaserScan** (topic: /scan), **Map** (topic: /map), and **Odometry** (topic: /odom).
- **TF Tree:** I observed the coordinate transforms, specifically the map → odom → base_footprint hierarchy, which is vital for understanding how local odometry is corrected by global mapping.
- **Map Saving:** After teleoperating the robot to explore the environment, the map was saved:
  ros2 run nav2_map_server map_saver_cli -f maps/my_map.



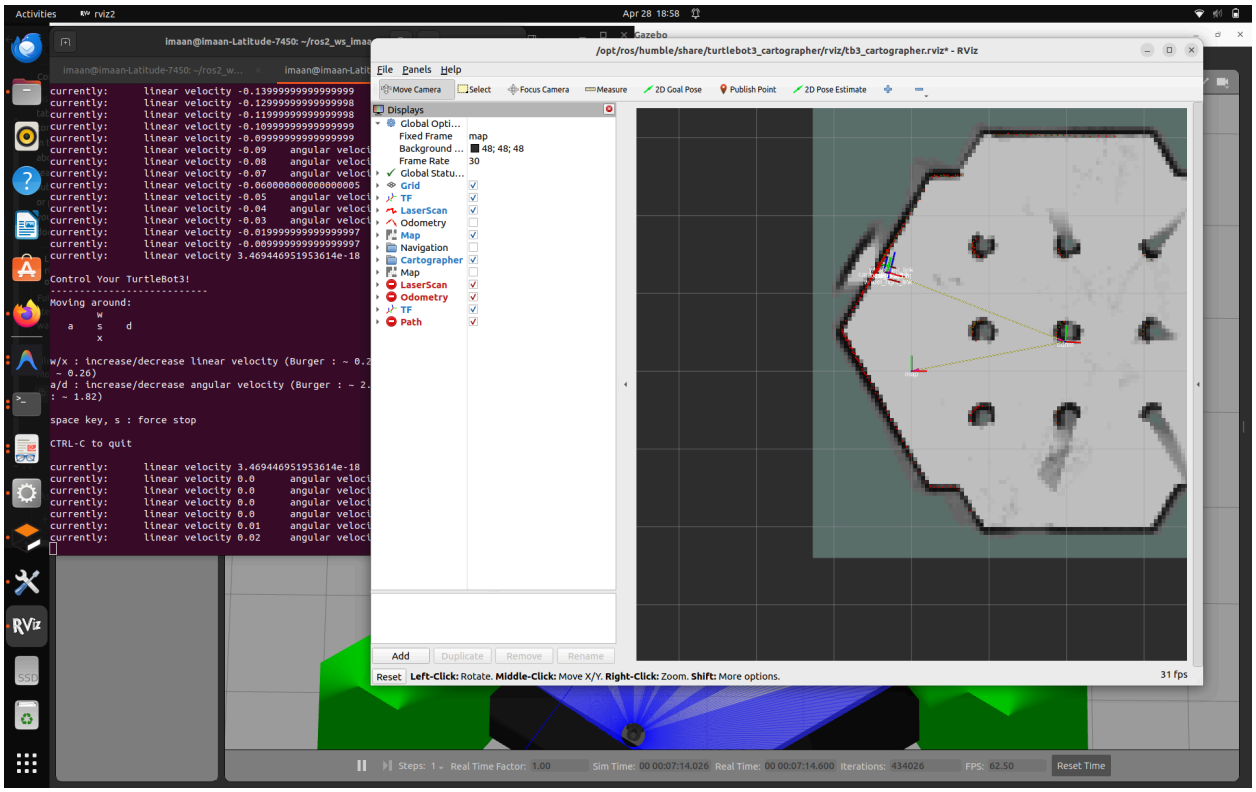
Phase III: Custom Node Development

I developed and registered two Python nodes in lab5_pkg to demonstrate automated control and data parsing.

1. **cmd_vel_publisher.py (Timer-Based Control):** * **Logic:** Uses a 2.0s timer to alternate the linear.x velocity between 0.2 m/s and 0.0 m/s.
    - **Purpose:** Demonstrates open-loop periodic motion control.
2. **odom_subscriber.py (Data Parsing):**
    - **Logic:** Subscribes to /odom and converts the quaternion orientation into a **Yaw angle (Degrees)** using math.atan2.
    - **Output:** Prints real-time position (x, y, z), linear velocity, and angular velocity to the terminal.


Activities
Terminal
Apr 28 18:58
imaan@imaan-Latitude-7450: ~/ros2_ws_imaan
imaan@imaan-Latitude-7450: ~/ros2_w...
imaan@imaan-Latitude-7450: ~/ros2_w...
currently: linear velocity -0.13999999999999999 angular velocity 0.0
currently: linear velocity -0.12999999999999998 angular velocity 0.0
currently: linear velocity -0.11999999999999998 angular velocity 0.0
currently: linear velocity -0.10999999999999999 angular velocity 0.0
currently: linear velocity -0.09999999999999999 angular velocity 0.0
currently: linear velocity -0.09 angular velocity 0.0
currently: linear velocity -0.08 angular velocity 0.0
currently: linear velocity -0.07 angular velocity 0.0
currently: linear velocity -0.060000000000000005 angular velocity 0.0
currently: linear velocity -0.05 angular velocity 0.0
currently: linear velocity -0.04 angular velocity 0.0
currently: linear velocity -0.03 angular velocity 0.0
currently: linear velocity -0.019999999999999997 angular velocity 0.0
currently: linear velocity -0.009999999999999997 angular velocity 0.0
currently: linear velocity 3.469446951953614e-18 angular velocity 0.0
Control Your TurtleBot3!

Moving around:
w
a s d
x
w/x : increase/decrease linear velocity (Burger : ~ 0.22, Waffle and Waffle Pi : ~ 0.26)
a/d : increase/decrease angular velocity (Burger : ~ 2.84, Waffle and Waffle Pi : ~ 1.82)
space key, s : force stop
CTRL-C to quit
currently: linear velocity 3.469446951953614e-18 angular velocity -0.1
currently: linear velocity 0.0 angular velocity 0.0
currently: linear velocity 0.0 angular velocity -0.1
currently: linear velocity 0.0 angular velocity -0.2
currently: linear velocity 0.0 angular velocity -0.30000000000000004
currently: linear velocity 0.01 angular velocity -0.30000000000000004
currently: linear velocity 0.02 angular velocity -0.30000000000000004
Gazebo
This version of Gazebo reaches end-of-life in January 2025. Consider migrating to the new Gazebo
Steps: 1
Real Time Factor: 1.00
Sim Time: 00 00:06:41.607
Real Time: 00 00:06:42.152
Iterations: 401607
FPS: 62.52
Reset Time


Activities
rviz2
Apr 28 18:58
imaan@imaan-Latitude-7450: ~/ros2_ws_imaan
imaan@imaan-Latitude-7450: ~/ros2_w...
imaan@imaan-Latit
currently: linear velocity -0.13999999999999999
currently: linear velocity -0.12999999999999998
currently: linear velocity -0.11999999999999998
currently: linear velocity -0.10999999999999999
currently: linear velocity -0.09999999999999999
currently: linear velocity -0.09 angular velocit
currently: linear velocity -0.08 angular velocit
currently: linear velocity -0.07 angular velocit
currently: linear velocity -0.060000000000000005
currently: linear velocity -0.05 angular velocit
currently: linear velocity -0.04 angular velocit
currently: linear velocity -0.03 angular velocit
currently: linear velocity -0.019999999999999997
currently: linear velocity -0.009999999999999997
currently: linear velocity 3.469446951953614e-18
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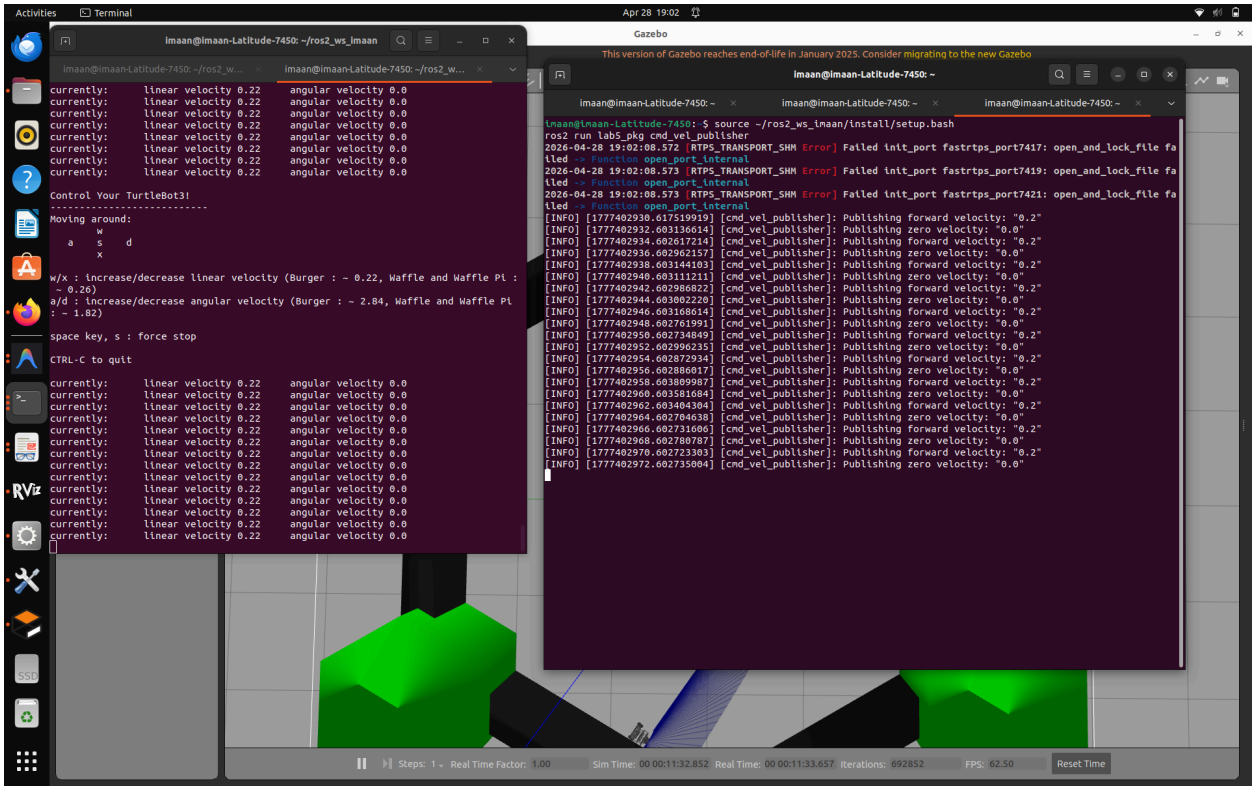
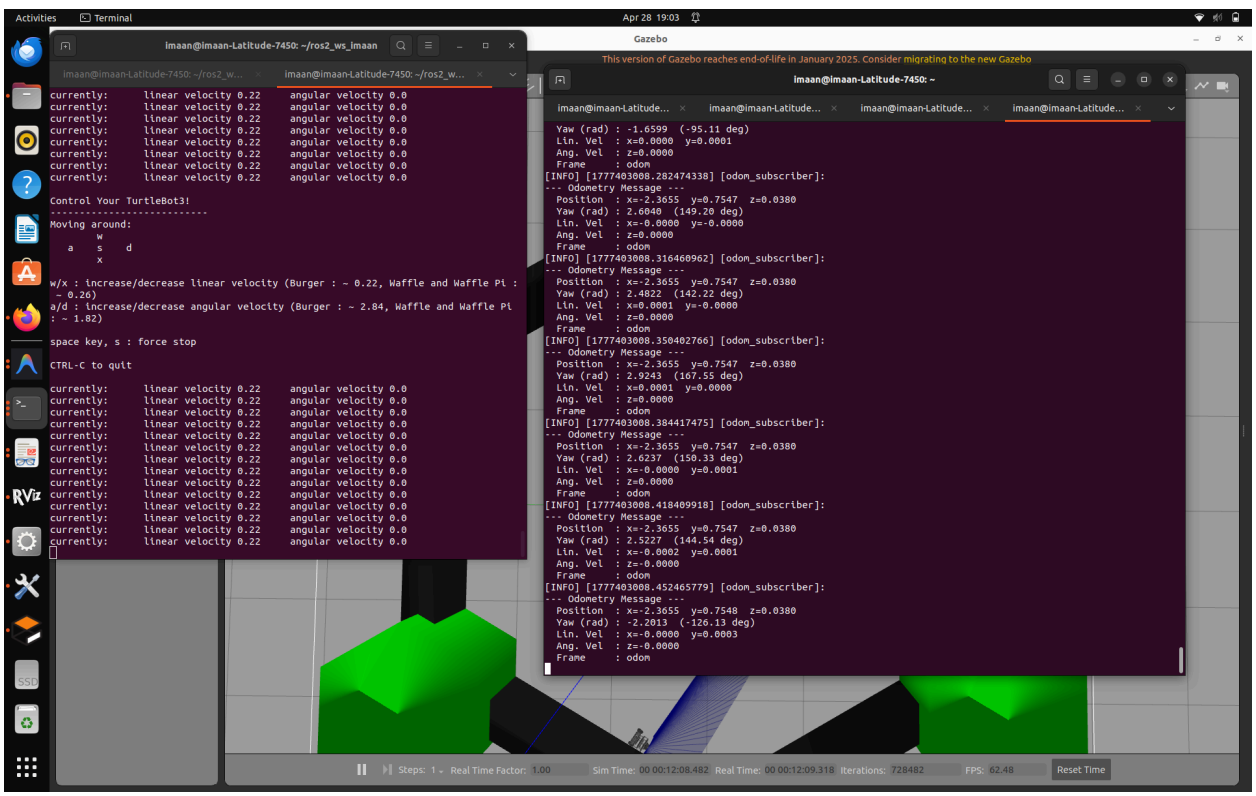
a s d
x
w/x : increase/decrease linear velocity (Burger : ~ 0.2
~ 0.26)
a/d : increase/decrease angular velocity (Burger : ~ 2.
: ~ 1.82)
space key, s : force stop
CTRL-C to quit
currently: linear velocity 3.469446951953614e-18
currently: linear velocity 0.0 angular velocit
currently: linear velocity 0.0 angular velocit
currently: linear velocity 0.0 angular velocit
currently: linear velocity 0.0 angular velocit
currently: linear velocity 0.01 angular velocit
currently: linear velocity 0.02 angular velocit
Gazebo
/opt/ros/humble/share/turtlebot3_cartographer/rviz/tb3_cartographer.rviz* - RViz
File Panels Help
Move Camera
Select
Focus Camera
Measure
2D Goal Pose
Publish Point
2D Pose Estimate
Displays
Global Opti...
Fixed Frame map
Background ... 48; 48; 48
Frame Rate 30
Global Statu...
Grid
TF
LaserScan
Odometry
Map
Navigation
Cartographer
Map
LaserScan
Odometry
TF
Path
Add
Duplicate
Remove
Rename
Reset
Left-Click: Rotate. Middle-Click: Move X/Y. Right-Click: Zoom. Shift: More options.
31 fps
Steps: 1
Real Time Factor: 1.00
Sim Time: 00 00:07:14.026
Real Time: 00 00:07:14.600
Iterations: 434026
FPS: 62.50
Reset Time


Activities
Terminal
Apr 28 19:02
Gazebo
This version of Gazebo reaches end-of-life in January 2025. Consider migrating to the new Gazebo
imaan@imaan-Latitude-7450: ~/ros2_ws_imaan
imaan@imaan-Latitude-7450: ~/ros2_w...
currently: linear velocity 0.22 angular velocity 0.0
Control Your TurtleBot3!

Moving around:
w
a s d
x
w/x : increase/decrease linear velocity (Burger : ~ 0.22, Waffle and Waffle Pi : ~ 0.26)
a/d : increase/decrease angular velocity (Burger : ~ 2.84, Waffle and Waffle Pi : ~ 1.82)
space key, s : force stop
CTRL-C to quit
imaan@imaan-Latitude-7450: ~
imaan@imaan-Latitude-7450: ~ $ source ~/ros2_ws_imaan/install/setup.bash
ros2 run lab5_pkg cmd_vel_publisher
2026-04-28 19:02:08.572 [RTPS_TRANSPORT_SHM Error] Failed init_port fastrtps_port7417: open_and_lock_file failed -> Function open_port_internal
2026-04-28 19:02:08.573 [RTPS_TRANSPORT_SHM Error] Failed init_port fastrtps_port7419: open_and_lock_file failed -> Function open_port_internal
2026-04-28 19:02:08.573 [RTPS_TRANSPORT_SHM Error] Failed init_port fastrtps_port7421: open_and_lock_file failed -> Function open_port_internal
[INFO] [1777402930.617519919] [cmd_vel_publisher]: Publishing forward velocity: "0.2"
[INFO] [1777402932.603136614] [cmd_vel_publisher]: Publishing zero velocity: "0.0"
[INFO] [1777402934.602617214] [cmd_vel_publisher]: Publishing forward velocity: "0.2"
[INFO] [1777402936.602902157] [cmd_vel_publisher]: Publishing zero velocity: "0.0"
[INFO] [1777402938.603144103] [cmd_vel_publisher]: Publishing forward velocity: "0.2"
[INFO] [1777402940.603111211] [cmd_vel_publisher]: Publishing zero velocity: "0.0"
[INFO] [1777402942.602986822] [cmd_vel_publisher]: Publishing forward velocity: "0.2"
[INFO] [1777402944.603002220] [cmd_vel_publisher]: Publishing zero velocity: "0.0"
[INFO] [1777402946.603168614] [cmd_vel_publisher]: Publishing forward velocity: "0.2"
[INFO] [1777402948.602761991] [cmd_vel_publisher]: Publishing zero velocity: "0.0"
[INFO] [1777402950.602734849] [cmd_vel_publisher]: Publishing forward velocity: "0.2"
[INFO] [1777402952.602996235] [cmd_vel_publisher]: Publishing zero velocity: "0.0"
[INFO] [1777402954.602872934] [cmd_vel_publisher]: Publishing forward velocity: "0.2"
[INFO] [1777402956.602886017] [cmd_vel_publisher]: Publishing zero velocity: "0.0"
[INFO] [1777402958.603809987] [cmd_vel_publisher]: Publishing forward velocity: "0.2"
[INFO] [1777402960.603581684] [cmd_vel_publisher]: Publishing zero velocity: "0.0"
[INFO] [1777402962.603404304] [cmd_vel_publisher]: Publishing forward velocity: "0.2"
[INFO] [1777402964.602704638] [cmd_vel_publisher]: Publishing zero velocity: "0.0"
[INFO] [1777402966.602731606] [cmd_vel_publisher]: Publishing forward velocity: "0.2"
[INFO] [1777402968.602780787] [cmd_vel_publisher]: Publishing zero velocity: "0.0"
[INFO] [1777402970.602723303] [cmd_vel_publisher]: Publishing forward velocity: "0.2"
[INFO] [1777402972.602735004] [cmd_vel_publisher]: Publishing zero velocity: "0.0"
Steps: 1
Real Time Factor: 1.00
Sim Time: 00 00:11:32.852
Real Time: 00 00:11:33.657
Iterations: 692852
FPS: 62.50
Reset Time


Activities
Terminal
Apr 28 19:03
Gazebo
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imaan@imaan-Latitude-7450: ~/ros2_ws_imaan
imaan@imaan-Latitude-7450: ~/ros2_w...
currently: linear velocity 0.22 angular velocity 0.0
Control Your TurtleBot3!

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a s d
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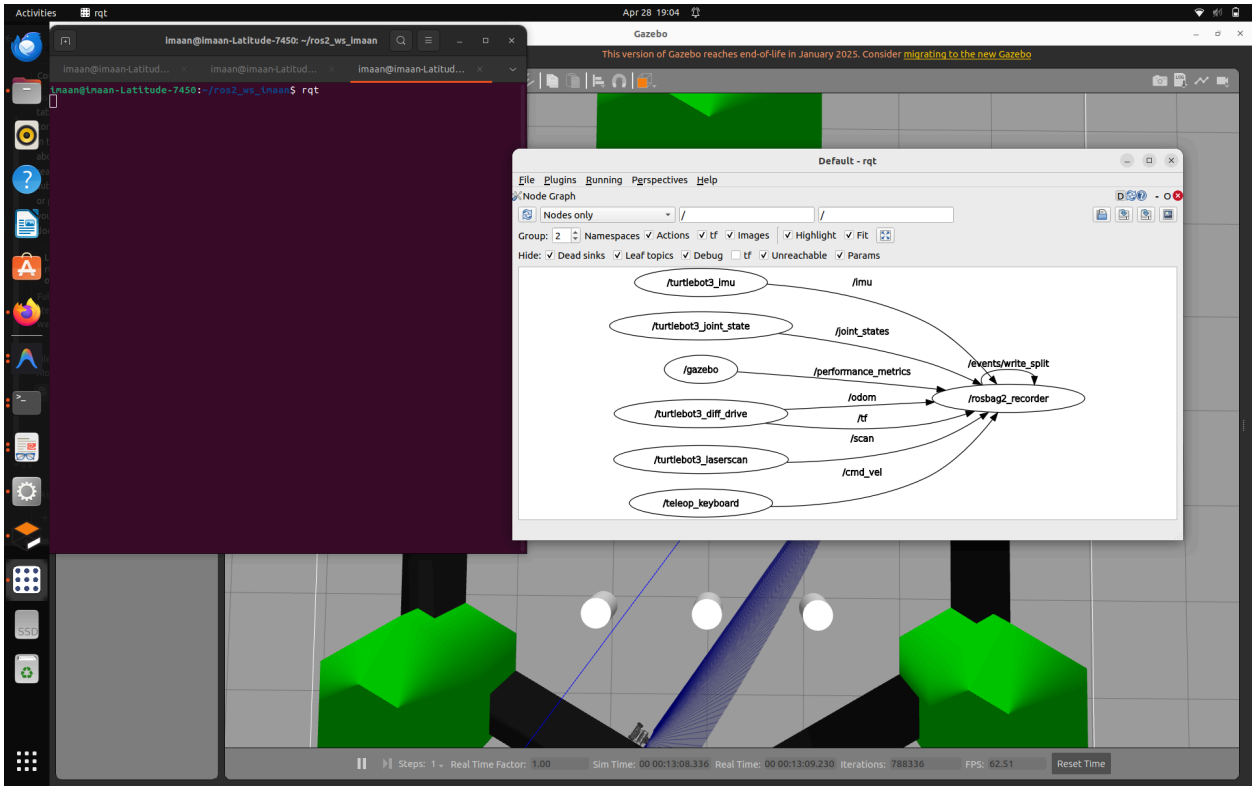
w/x : increase/decrease linear velocity (Burger : ~ 0.22, Waffle and Waffle Pi : ~ 0.26)
a/d : increase/decrease angular velocity (Burger : ~ 2.84, Waffle and Waffle Pi : ~ 1.82)
space key, s : force stop
CTRL-C to quit
imaan@imaan-Latitude-7450: ~
Yaw (rad) : -1.6599 (-95.11 deg)
Lin. Vel : x=0.0000 y=0.0001
Ang. Vel : z=0.0000
Frame : odom
[INFO] [1777403008.282474338] [odom_subscriber]:
--- Odometry Message ---
Position : x=-2.3655 y=0.7547 z=0.0380
Yaw (rad) : 2.6040 (149.20 deg)
Lin. Vel : x=-0.0000 y=-0.0000
Ang. Vel : z=0.0000
Frame : odom
[INFO] [1777403008.316460962] [odom_subscriber]:
--- Odometry Message ---
Position : x=-2.3655 y=0.7547 z=0.0380
Yaw (rad) : 2.4822 (142.22 deg)
Lin. Vel : x=0.0001 y=-0.0000
Ang. Vel : z=0.0000
Frame : odom
[INFO] [1777403008.350402766] [odom_subscriber]:
--- Odometry Message ---
Position : x=-2.3655 y=0.7547 z=0.0380
Yaw (rad) : 2.9243 (167.55 deg)
Lin. Vel : x=0.0001 y=0.0000
Ang. Vel : z=0.0000
Frame : odom
[INFO] [1777403008.384417475] [odom_subscriber]:
--- Odometry Message ---
Position : x=-2.3655 y=0.7547 z=0.0380
Yaw (rad) : 2.6237 (150.33 deg)
Lin. Vel : x=-0.0000 y=0.0001
Ang. Vel : z=0.0000
Frame : odom
[INFO] [1777403008.418409918] [odom_subscriber]:
--- Odometry Message ---
Position : x=-2.3655 y=0.7547 z=0.0380
Yaw (rad) : 2.5227 (144.54 deg)
Lin. Vel : x=-0.0002 y=0.0001
Ang. Vel : z=-0.0000
Frame : odom
[INFO] [1777403008.452465779] [odom_subscriber]:
--- Odometry Message ---
Position : x=-2.3655 y=0.7548 z=0.0380
Yaw (rad) : -2.2013 (-126.13 deg)
Lin. Vel : x=-0.0000 y=0.0003
Ang. Vel : z=-0.0000
Frame : odom
Steps: 1
Real Time Factor: 1.00
Sim Time: 00 00:12:08.482
Real Time: 00 00:12:09.318
Iterations: 728482
FPS: 62.48
Reset Time



3. Observations & Technical Analysis

Motion Discrepancies

Feature	Expected	Observed	Technical Reason
Return to Origin	$(0, 0, 0)$	Small offset (~2–5 cm)	Accumulation of wheel slip and integration drift.
Path Straightness	Perfectly straight	Slight curve	Slight asymmetries in the differential drive physics model.
LiDAR Data	Solid clean lines	Minor noise/jitter	Gazebo's sensor noise model simulating real-world LDS-01 interference.

The TF Hierarchy

Understanding the Transform (TF) tree was a key takeaway.

- The odom frame represents the robot's starting position.
- The map frame is the global fixed reference.
- The transform from map to odom represents the "drift correction" provided by the SLAM algorithm.

4. Conclusion

This lab effectively bridged the gap between code and physical behavior. By using **Rosbag** to record all active topics (ros2 bag record -a), I captured the entire experimental run for later playback and analysis. The primary challenge noted was **odometry drift**, which emphasized why Mechatronics engineers must rely on sensor fusion (LiDAR + IMU + Encoders) via SLAM rather than simple dead-reckoning for reliable robot navigation.

Deliverables Included:

- **Code:** cmd_vel_publisher.py and odom_subscriber.py.
- **Data:** my_map.pgm (Saved SLAM Map) and ros2 bag recording.
- **Visuals:** Screenshots of rqt_graph and RViz sensor plugins.